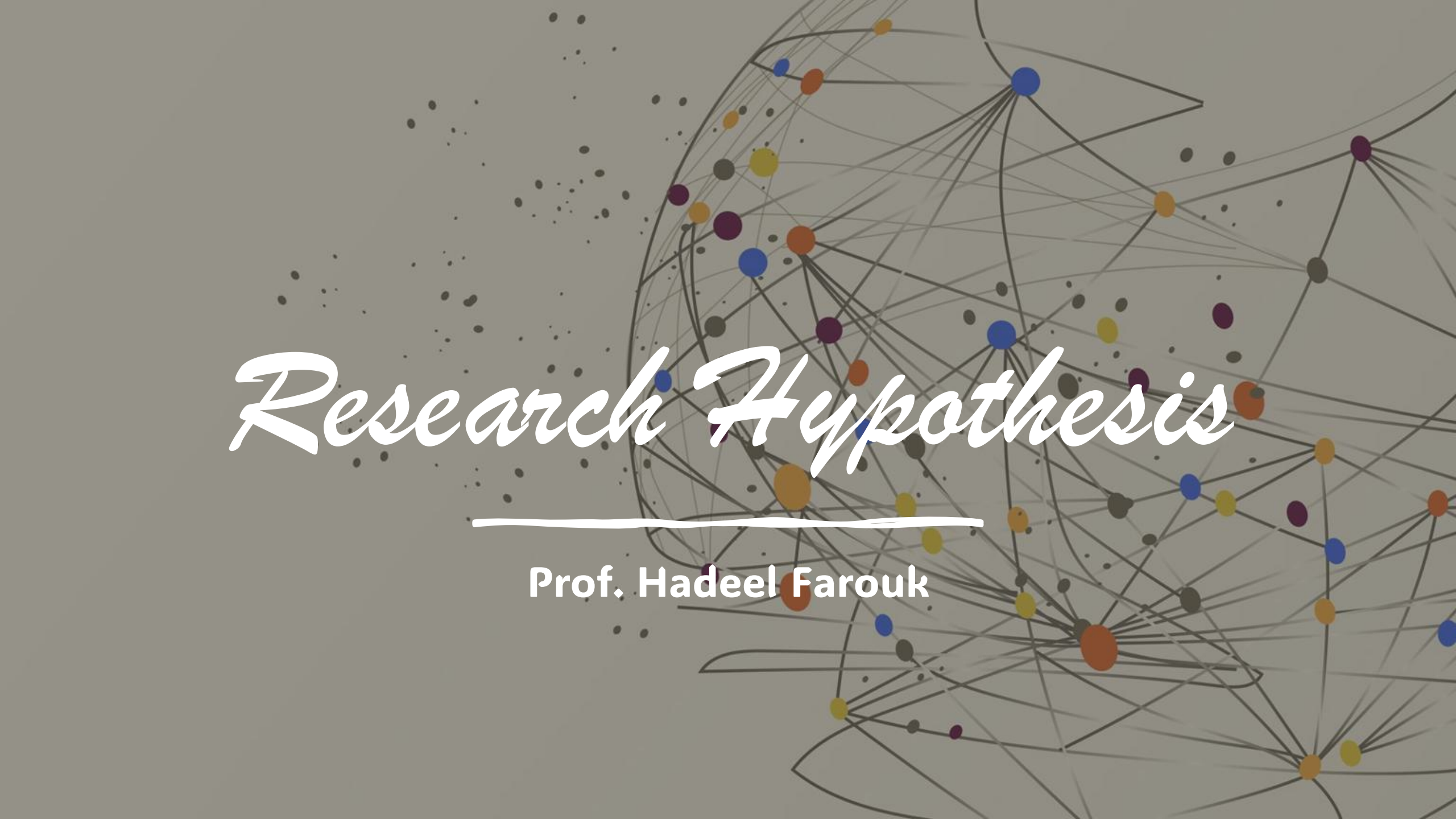


Research Hypothesis

Prof. Hadeel Farouk

“Medicine is the science of uncertainty and the art of probability.”

Sir William Osler (1849–1919)

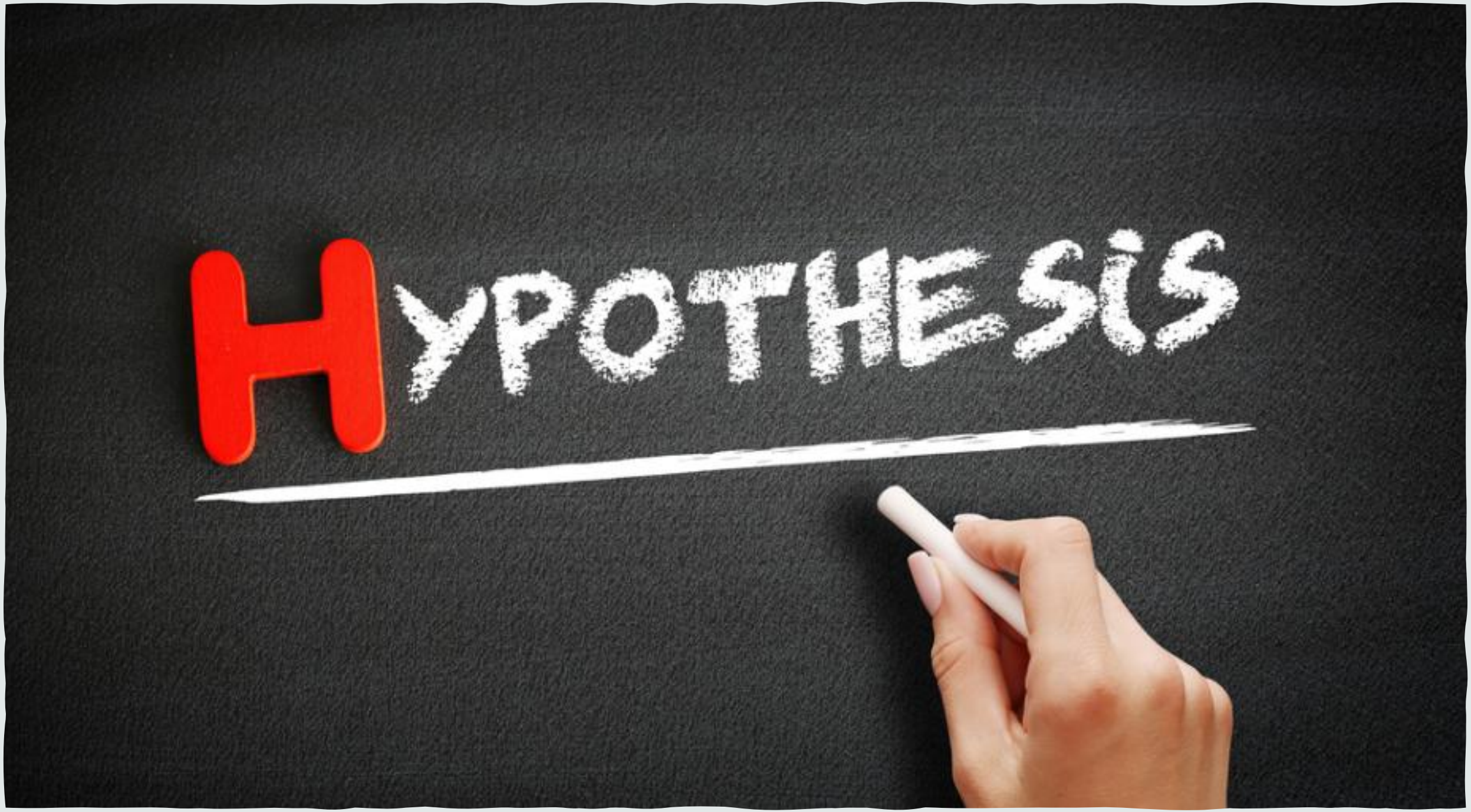
Research Question

is what a study aims to answer after data analysis and interpretation. The answer is written in length in the discussion section of the paper.

Answer

Research Hypothesis

It is an educated statement of an expected outcome. This statement is based on background research and current knowledge. The research hypothesis makes a specific prediction about a new phenomenon or a formal statement on the expected relationship between an independent variable(What you change)and a dependent variable(What you measure).


HYPOTHESIS

What is a hypothesis?

The sole purpose of a hypothesis is to predict your paper's findings, data, and conclusion. It comes from a place of curiosity.

When you write a hypothesis, you're essentially making an educated guess based on scientific prejudices and evidence, which is further proven or disproven through the scientific method.

What is a hypothesis?

For some research projects, you might have to write several hypotheses that address different aspects of your research question.

A hypothesis is not just a guess

- It should be based on existing theories and knowledge.
- It also must be testable, which means you can support or disprove it through scientific research methods (such as experiments, observations, and statistical data analysis).

Variables in hypotheses

Hypotheses propose a relationship between two or more types of variables.

- An independent variable is something the researcher changes or controls.
- A dependent variable is something the researcher observes and measures.

. **Phrase your hypothesis in three ways**

To identify the variables, you can write a simple prediction in *if…then* form.

The **first part** of the sentence states the independent variable, and the **second part** states the dependent variable.

EXAMPLE

If a first-year student starts attending more lectures, then their exam scores will improve.

The number of lectures attended by first-year students has a positive effect on their exam scores.

- **If you are comparing two groups, the hypothesis can state what difference you expect to find between them.**

First-year students who attended most lectures will have better exam scores than those who attended few lectures.

Write a null hypothesis

If your research involves statistical hypothesis testing, you will also have to write a null hypothesis. The null hypothesis is the default position that there is no association between the variables. The null hypothesis is written as H_0, while the alternative hypothesis is H_1 or H_a.

- **H_0: The number of lectures attended by first-year students has no effect on their final exam scores.**

- **H_1: The number of lectures attended by first-year stud ents has a positive effect on their final exam scores**

The Scenario: A researcher wants to know if going to lectures helps first-year students get better grades.

- **Independent Variable: Number of lectures attended.**
- **Dependent Variable: Final exam scores.**



The Hypotheses:

- **H_0 (Null): "The number of lectures attended by first-year students has no effect on their final exam scores."**
Going to class makes zero difference. Any difference in scores is just due to random chance.
- **H_1 (Alternative): "The number of lectures attended by first-year students has a positive effect on their final exam scores."**
Going to class actually helps students score higher.

You calculate the probability (p-value) of seeing your data if the Null Hypothesis were true.

- If the probability is very low (usually less than 5%), you reject the Null Hypothesis and say, "The evidence supports the Alternative Hypothesis.“

- If the probability is high, you fail to reject the Null Hypothesis (meaning, there isn't enough evidence to prove an effect).

A Null Hypothesis EXAMPLE

An investigator might examine three types of reinforcement for children with autism: verbal cues, a reward, and no reinforcement. The investigator collects behavioral measures to assess the children's social interaction with their siblings.

A null hypothesis might read H_0,

There is no significant difference between the effects of verbal cues, rewards, and no reinforcement in terms of social interaction for children with autism and their siblings

Hypothesis examples

Research question	Hypothesis	Null hypothesis
What are the health benefits of eating an apple a day?	Increasing apple consumption in over-60s will result in decreasing frequency of doctor's visits.	Increasing apple consumption in over-60s will have no effect on frequency of doctor's visits.
Which airlines have the most delays?	Low-cost airlines are more likely to have delays than premium airlines.	Low-cost and premium airlines are equally likely to have delays.
Can flexible work arrangements improve job satisfaction?	Employees who have flexible working hours will report greater job satisfaction than employees who work fixed hours.	There is no relationship between working hour flexibility and job satisfaction.
What effect does daily use of social media have on the attention span of under-16s?	There is a negative correlation between time spent on social media and attention span in under-16s.	There is no relationship between social media use and attention span in under-16s.

Research hypothesis checklist

Here is a 10-point checklist for a good research hypothesis:

1. **Testable**: A research hypothesis should be able to be tested via experimentation or observation.
2. **Specific**: A research hypothesis should clearly state the relationship between the variables being studied.
3. **Based on prior research**: A research hypothesis should be based on existing knowledge and previous research in the field.
4. **Falsifiable**: A research hypothesis should be able to be disproven through testing.
5. **Clear and concise**: A research hypothesis should be stated clearly and concisely.

Research hypothesis checklist

6. **Logical**: A research hypothesis should be logical and consistent with the current understanding of the subject.
7. **Relevant**: A research hypothesis should be relevant to the research question and objectives.
8. **Feasible**: A research hypothesis should be feasible to test within the scope of the study.
9. **Reflects the population**: A research hypothesis should consider the population or sample being studied.
10. **Uncomplicated**: A good research hypothesis is written in a way that is easy for the target audience to understand.

ALTERNATIVE HYPOTHESIS

NULL HYPOTHESIS

COMPLEX HYPOTHESIS



SIMPLE HYPOTHESIS

STATISTICAL HYPOTHESIS

EMPIRICAL HYPOTHESIS

TYPE OF HYPOTHESIS

Simple Hypothesis:

- A simple Hypothesis is a Hypothesis that reflects or shows a Relationship between Two Variables- Independent and Dependent Variables.

For Example,

- The higher the unemployment, the higher would be the rate of crime in society.

Complex Hypothesis:

- A Complex Hypothesis is a Hypothesis that reflects a relationship ...more than Two Variables.

For example,

- The higher the Poverty, the higher the illiteracy (الأمية) in society, higher will be the rate of crime(Three Variables - Two Independent and One Dependent)

Alternative Hypothesis:

Statistical Hypothesis:

- A Hypothesis, that can be verified statistically, is known as a Statistical Hypothesis.

Directional Hypothesis

A directional hypothesis predicts the positivity or negativity of the effect of an intervention before the test is conducted.

Non-Directional Hypothesis

A non-directional hypothesis does not specify the predicted direction (e.g. positivity or negativity) of the effect of the independent variable on the dependent variable.

These hypotheses predict an effect but stop short of saying what that effect will be.

Example 1:

Hypothesis for a study on the effects of exercise on weight loss:

Null Hypothesis (H0): There is no significant difference in weight loss between individuals who engage in regular exercise and those who do not.

Alternative Hypothesis (H1): Individuals who engage in regular exercise will experience greater weight loss compared to those who do not exercise.

Example 2:

Hypothesis for a study on the impact of social media on self-esteem:

Null Hypothesis (H0): There is no significant relationship between social media usage and self-esteem levels.

Alternative Hypothesis (H1): Increased social media usage is associated with lower self-esteem levels.

Example 3:

Hypothesis for a study on the effectiveness of a new teaching method in improving student performance:

Null Hypothesis (H0): There is no significant difference in student performance between the traditional teaching method and the new teaching method.

Alternative Hypothesis (H1): The new teaching method leads to improved student performance compared to the traditional teaching method.


Thank
You